Design and Evaluation of a Reliable Data Engineering and Analytics Platform for Finnish District Heating

A Design Science Research project with a balanced data analyst and data engineer competency pathway

Project	Revised proposal
Student / programme	Li Yu · Data Expertise and Artificial Intelligence · TAMK Master's
Revision / status	14 October 2026 · Draft for author and supervisor review; approval remains pending
Proposed period	November 2026–May 2027; official milestones to be agreed in Wihi
Capacity / starting level	20 hours per week; strong SQL, Python and Power BI skills, self-reported by the author
Career objective	Demonstrable applied competence for data analyst, BI developer, data engineer and related analytics engineer roles in Finland
Working-life partner	To be confirmed; an independently proposed topic requires programme approval

Executive summary

The thesis will design, implement and evaluate a reproducible platform integrating public Helen district-heating data and Finnish Meteorological Institute weather observations. Its core contribution is evidence about pipeline reliability, data provenance and the correctness and usability of analytical measures. It will connect software engineering with statistical interpretation and clear communication of findings.

The local reference implementation will use Python, PostgreSQL, Docker, SQL/dbt and Power BI. A bounded Microsoft Fabric deployment will demonstrate cloud orchestration and Delta/PySpark processing if capacity is available. A single declared environment will be used for the principal research experiments; local and cloud measurements will be reported separately.

A 560-hour planning budget, including 60 hours of contingency, is proposed within the existing seven-month window. Two research questions remain mandatory. Forecasting is a stretch experiment. The accompanying competency matrix defines what must be built, tested and explained before each skill is claimed in a portfolio.

The intended result is strong evidence of a coherent set of transferable skills. One thesis cannot establish mastery of every platform, replace professional experience or guarantee employment. Remaining gaps are explicitly identified for later learning.

1. Problem, purpose and research questions

1.1 Problem and working-life relevance

Public heating and weather data can differ in schema, interval, measurement meaning and publication history. Unchecked parsing, duplicated hours and inconsistent report filters can produce plausible but irreproducible findings. The practical problem is to establish a trustworthy chain from source observations to an explanation that an analyst or reviewer can inspect.

The proposed users are an energy analyst investigating demand patterns and a data engineer maintaining the pipeline. They are design personas until validated with a reviewer; no utility partnership or operational requirement is assumed. The project retains the domain of the previous MicroStation/hydraulic proposal while replacing its implementation focus with open-data engineering and business analytics.

1.2 Purpose and objectives

The purpose is to evaluate how explicit data contracts, repeatable transformations and governed measures support reliable analysis of district-heating time series. The research contribution will be a tested artefact and an evidence-based account of its design choices, limitations and transferability to similar small analytical systems.

- O1: integrate two sources with immutable snapshots, documented units, provenance and recoverable incremental processing.
- O2: implement a tested dimensional model and reconcile five analytical measures across SQL and Power BI.
- O3: answer predefined demand, weather and data-quality questions using defensible analysis and a concise decision memo.
- O4: evaluate reliability, reproducibility, performance and task completion using a frozen protocol and retained results.
- O5: produce independently reproducible engineering and analyst portfolio evidence, including a limited cloud demonstration when access permits.

Question	Research question and answer evidence
RQ1 · engineering	To what extent does the designed pipeline meet predefined requirements for completeness accounting, traceability, idempotency and recovery when integrating Helen and FMI data? Evidence: controlled fixtures, replay and revision tests, fault injection, run logs and reproducibility checks.
RQ2 · analytics	To what extent do the dimensional and semantic models reproduce independently calculated measures and support predefined district-heating analysis tasks? Evidence: SQL–DAX reconciliation, a statistical case study, task results and bounded reviewer feedback.

Improvement claims require a fair, documented baseline. These questions remain answerable if only absolute properties can be measured. Forecasting will not become a third required research question unless the supervisor approves a revised scope.

2. Research method and knowledge base

2.1 Design Science Research

The study will follow the problem–design–demonstration–evaluation–communication logic of Design Science Research [A1]. Evaluation will be planned early and refined through formative checks before the final experiments [A2]. Successful implementation alone is insufficient: the report must connect requirements, design decisions, measurements and answers to RQ1/RQ2.

Stage	Method and retained evidence
Problem and objectives	Review literature and source constraints; define personas and five analytical tasks; document reviewer input separately from author assumptions.
Design	Map each requirement to an implementation and a test. Record decisions on grain, units, revisions, weather alignment, SQL transformations and deployment.
Construction	Develop small vertical increments: source → validated table → measure → visual. Maintain versioned code, change reviews, tests and a research diary.
Demonstration	Run a frozen historical period and show demand seasonality, weather association and data completeness. Reproduce one source revision and one recovery scenario.
Evaluation	Run E1–E7 in Sections 6–7. Retain failures as well as successes. Compare with a minimal baseline only where the same task and input can be used.
Communication	Write the academic report, technical handover and analyst memo. Present two portfolio narratives based on the same verified artefact.

2.2 Literature and evidence strategy

Begin with DSR and evaluation literature [A1–A2], data quality from the consumer perspective [A3], dimensional modelling and semantic-model design [T1], and research on energy/weather time series. Search the TAMK library, Scopus or comparable accessible academic indexes, and follow relevant references. Aim for 15–20 carefully selected scholarly sources, retaining foundational work alongside recent studies; this is a planning target, not an institutional requirement.

Maintain a literature matrix with question, method, dataset, findings, limitations and design implication. Record search terms, databases, dates and inclusion reasons. Prioritise work on reproducibility, missingness, temporal alignment and analytical validity. Establish the precise literature gap after review; do not claim this platform is novel merely because the selected tools are combined.

Use vendor documentation for product behaviour and employer pages for career context, not as substitutes for research evidence. The four employer examples in Appendix B form an illustrative sample, not a systematic Finnish labour-market study.

2.3 Baseline and limits of inference

Create a small fixed-file Pandas/SQL baseline that calculates the same five measures and records manual steps. Treat it as a reproducible project baseline, not an existing Helen workflow. If it cannot be made comparable within the budget, report absolute performance and correctness without claiming time savings. Observational weather associations will not be interpreted as causal effects or operational savings.

3. Data scope, contracts and model

Core scope: one aggregated heating system, two publishers, a frozen overlapping period, three report pages and five governed measures. Target 24 consecutive months if both sources support them; otherwise use at least 12 months covering seasons and daylight-saving transitions and disclose the reduced comparisons. Freeze the period after source audit. Full-history profiling may inform selection without expanding every experiment.

Source	Contract and feasibility decisions
Helen CSV [D1]	Retain the original file, URL, retrieval date, checksum, licence and schema. The local audit recorded 91,767 rows and columns datetime_utc and dh_MWh. The webpage calls the series heating power, so reconcile field units and interval meaning before deriving energy or power KPIs.
FMI weather [D2]	Verify a repeatable WFS observation extract over the chosen period. Specify station ID, coordinates, temperature parameter, unit, UTC convention, interval and quality codes. Preselect one representative Helsinki station; justify any fallback station and test a small sensitivity sample.
Source revision	Version changed snapshots and record observation time separately from retrieval time. Distinguish scheduled checks from actual publisher updates. Demonstrate incremental processing with historical partitions when current updates are unavailable.

3.1 Parsing and temporal rules

The observed Helen sample uses semicolon separation, decimal commas and day-first timestamp text. Record explicit parsing rules and fail on unexpected schema or values. A column name containing “utc” is evidence to verify, not sufficient proof of timestamp semantics. Store original text, original unit, UTC interval start/end, source hash, row identity and ingestion run ID.

Use UTC for keys and Europe/Helsinki for reporting. Build expected intervals from a calendar spine. Test 23-hour and 25-hour local days, leap days, interval boundaries and duplicate local clock times. Do not silently impute missing demand; keep the gap and its quality status. Weather interpolation, if used, must be bounded, flagged and compared with an unfilled analysis.

3.2 Proposed analytical model

Table / layer	Declared grain and responsibility
Raw / Bronze	Immutable source files or responses plus manifests; retain all revisions and ingestion metadata.
Validated / Silver	Typed observations with explicit keys, units, validation outcomes and rejected-record reasons.
fact_heat_hour	One heating system × UTC hour; confirmed energy or power field, interval duration, source reference and quality status.
fact_weather_hour	One station × UTC hour for the selected temperature measure; count/coverage of underlying observations.
Gold / reporting	Dimensions for date, hour, system and station; a validated one-to-one aligned heat/weather view after station selection. Separate pipeline-run and quality summaries.

Specify keys, nullability, referential integrity and revision rules. Test joins against row multiplication. Keep independent facts separate in the semantic model; use conformed dimensions or the explicitly aligned view. Add slowly changing dimensions only if a real metadata change requires them.

4. Engineering design and cloud boundary

Flow	Implementation and evidence
Acquire → preserve	Python CLI adapters for CSV and WFS; bounded requests, timeouts, retry/backoff, immutable raw responses and manifest checksums.
Validate → load	Pandas parsing, explicit contracts, quarantine, transaction boundaries, atomic publication and a run ledger. PostgreSQL provides the local reference store.
Transform → model	Versioned SQL models in dbt Core with the PostgreSQL adapter; dependency graph, model documentation, uniqueness, relationship and custom quality tests [T2].
Model → analyse	Power BI semantic model, documented DAX/Power Query and parameterised connections; Python analysis reads curated outputs.
Operate → verify	Docker Compose, pinned dependencies, migrations, pytest, CI with synthetic fixtures, logs, a runbook and release evidence.

4.1 Reliability and operational practice

Use source revision plus business key to distinguish retries from corrections. A repeat of the same batch must leave curated records unchanged. A revised batch must update only the intended records and retain lineage. Use transactions and a publish step so an interrupted load does not expose partial analytical output. Define backfill commands and checkpoints; test them with controlled failures.

Record run ID, source revision, configuration, Git commit, stage row counts, rejects, retry count, duration and final status. Separate source coverage lag from time since successful ingestion. Provide a bounded scheduler and dependency sequence; simulate timeouts in tests rather than repeatedly disrupting public services.

Separate the loader and read-only reporting roles. The current local superuser is a setup convenience, not the final reporting account. Test denied writes, secret handling and restore into a separate test database. Add one measured SQL optimisation with an unchanged-result check; report when a simpler design is sufficient for this small dataset.

4.2 Microsoft Fabric work package · maximum 50 hours

Preferred cloud demonstration: one workspace, one Data Factory pipeline, one Lakehouse with raw/validated/analytical Delta tables, one focused PySpark notebook and a Power BI model. Demonstrate a scheduled run, rerun, failure recovery, access configuration and matching results on the same frozen sample. Export code, settings, run evidence and a cost/capacity note. Avoid implementing a second full warehouse stack.

Confirm institutional capacity and sharing rights in November. If only a trial is available, start it close to the planned March demonstration; current documentation describes a 60-day trial, subject to availability [T3]. Select the principal evaluation environment before final experiments and preserve its configuration. Keep the local reference executable regardless of access.

If cloud access fails, complete RQ1/RQ2 locally and label Fabric/PySpark as an unmet practical skill until a real deployment is demonstrated. Architecture diagrams alone do not establish hands-on cloud competence. No paid subscription is assumed in the budget.

5. Analyst work, measures and communication

5.1 Analytical questions and outputs

The analyst strand will produce explanations and recommendations as well as a dashboard. Questions are fixed before final analysis: when are demand and peaks concentrated; how do comparable periods differ; how is demand associated with weather after seasonal/calendar adjustment; which conclusions are sensitive to incomplete data; and can a reviewer trace a reported result to its inputs?

Measure	Definition to freeze and validate
M1 · period energy	Sum validated interval energy in MWh. If the source is average MW, integrate using verified interval duration. If it is instantaneous power, do not claim energy without a justified integration method.
M2 · peak demand	Maximum validated interval-average MW and its UTC/local timestamp. Derive MW from MWh ÷ interval hours only when the source represents interval energy; disclose tied peaks.
M3 · comparable change	Percentage change between matched complete periods; return blank/flag for an invalid or zero denominator. Match calendar coverage and display completeness alongside the change.
M4 · data completeness	Accepted distinct expected-hour keys ÷ all expected-hour keys in the selected system/period. Track source absence, rejection and join loss separately; weather has its own denominator.
M5 · demand–temperature relationship	A reproducible summary of aligned daily demand by predeclared temperature bands, with sample sizes and missingness; accompany it with the statistical case study below.

Each measure has an owner (the thesis author unless agreed otherwise), business definition, unit, grain, filter rules, SQL reference, DAX implementation, missing-data policy and tolerance. A separate freshness indicator shows the newest source observation and last successful ingestion rather than implying that archived data is live.

5.2 Statistical case study

Use exploratory distributions, seasonal/day-of-week summaries and temperature–demand plots. Fit a small interpretable daily model using temperature and calendar terms, with a documented seasonal adjustment. Compare with a calendar-only baseline on a chronological holdout. Inspect residuals and autocorrelation; use a justified block-bootstrap or time-series-aware uncertainty method rather than treating hourly rows as independent samples. Report effect sizes, uncertainty and practical interpretation. Compare complete cases with any approved weather-fill rule.

This is a historical explanatory analysis. Heating-degree-day features may be added after documenting the base-temperature convention. A next-day forecast is optional: seasonal-naive versus one simple model, rolling-origin validation, MAE/RMSE, and only predictors available at issue time. Future observed weather may support a labelled scenario, never a claimed operational forecast.

5.3 Decision support deliverables

Build three Power BI pages: demand and comparable periods; weather and patterns; quality and lineage. Test slicers, drill-through, tooltips, readable units, accessible colours and keyboard navigation. Deliver a small Excel/Power Query audit workbook and a two-page memo with three supported findings, uncertainty, proposed follow-up actions and the evidence needed before operational use. Revenue, cost savings and emissions claims require additional data and are outside the core scope.

6. Evaluation protocol: engineering

Freeze input hashes, expected outcomes, thresholds and experiment scripts before final evaluation. Retain genuine source gaps; use clearly labelled synthetic fixtures for known defects and edge cases. The thresholds below are proposed acceptance criteria, not completed results.

Experiment	Controlled procedure	Acceptance / retained evidence
E1 · parsing, coverage and lineage RQ1	Use a normal week, winter week, both DST transitions and targeted fixtures for malformed values, duplicates, missing hours and changed columns. Compare input/output keys with an independently prepared expected set.	All injected defects are detected or explicitly rejected; zero unexplained accepted-key loss or join multiplication; every curated record traces to a source revision and run. Report actual source completeness rather than demand 100% coverage.
E2 · replay and correction RQ1	Run the same fixed batch twice, then a controlled corrected snapshot and overlapping backfill. Compare keyed values and provenance before/after.	Same revision: identical curated keys and values, zero duplicate business keys. Correction: only expected values change and both revisions remain traceable. Audit timestamps may differ without invalidating content equality.
E3 · fault and recovery RQ1	Inject a transient HTTP failure, missing input, schema change and interruption before publication. Then rerun after removing the fault.	No partial Gold publication; bounded retries; actionable failed status; recovery matches a clean reference run. Record time to diagnosis/recovery and any required manual steps.
E4 · reproducibility and operations RQ1	From a clean checkout, start containers, load a licensed or synthetic fixture and run tests. Restore a backup into a separate test database and check reporting-role permissions.	Documented commands reproduce expected output. Read-only role cannot modify data. Restored keys, values and audit records match. Capture dependency versions, image version, commit and command output.

Performance sub-study

On the declared environment, measure full-load and incremental elapsed time and rows per second on the same frozen input. Run five measured repetitions after any documented warm-up; report median and range with hardware, cache state and capacity. Compare one indexing/query/materialisation change with its unoptimised form, verifying identical results. If costs are estimated, disclose the assumptions and separate them from actual billed amounts.

The real dataset is modest. A larger deterministic synthetic fixture may reveal bottlenecks, but it cannot justify claims about terabyte-scale production operation. CI tests use small fixtures and avoid live public API dependencies; a separate controlled integration check verifies source access.

7. Evaluation protocol: analytics and validity

Experiment	Controlled procedure	Acceptance / retained evidence
E5 · metric correctness RQ2	For each M1–M5, compare independently written SQL and DAX over at least six filter cases: full period, month, day, empty selection, missing-data period and DST boundary. Include hand-calculated tiny fixtures.	All counts/keys match exactly. For numeric measures, proposed tolerance is $\|DAX - SQL\| \le \max(0.001 \text{ unit}, 10^{-6} \times \|SQL\|)$, with percentages in percentage points. Justify a different tolerance before testing. No unexplained mismatch remains.
E6 · analytical tasks RQ2	Use five tasks: identify a peak; compare matched periods; explain a weather pattern; diagnose missing coverage; trace one KPI to source. Invite 2–3 reviewers, aiming for analyst/domain and technical perspectives.	Compare answers with a reference rubric. Record correctness, time, assistance and clarity comments. Target at least 4/5 tasks completed without assistance per reviewer and correct answers for all five after issues are resolved. Retain initial and retest results separately.
E7 · analysis reproducibility RQ2	Recreate the notebook, model, charts and memo from frozen Gold data. Verify split dates, assumptions, missingness policy and a weather/station sensitivity check.	Reported values and charts regenerate; uncertainty and confounding are discussed; all three memo findings have a query/chart and a limitation. A null or weak association is a valid result.

7.1 Reviewer design and fallback

Confirm access, consent and any programme-specific requirements before involving reviewers. Use the same task wording, anonymised IDs and expected-answer rubric. Collect only necessary feedback and timings. If no reviewers are available, use a scripted walkthrough and describe it as author verification; it cannot support a claim of usability for real users.

A small convenience sample supports formative insight, not population-level usability estimates. If a manual baseline task comparison is attempted, counterbalance task order where feasible and report learning effects. Independent KPI queries and hand-calculated fixtures reduce the risk that the same implementation error appears in every check.

7.2 Validity and contribution

Internal validity: freeze transformations before the final run, document changes, compare equivalent periods and separate source defects from pipeline defects. Construct validity: define reliability, completeness and task success operationally. External validity: limit conclusions to this system, period, source pair and evaluation environment; discuss what would change for other organisations or larger workloads.

The report will answer each research question with a requirement–design–test–result chain, explain failed criteria and distil a small set of design lessons. Proposed lessons are not conclusions until supported by the measurements. A portfolio competency is a separate learning outcome, not proof of scientific novelty or professional seniority.

8. Workload, timetable and decision gates

Planning assumption agreed with the author: 20 hours per week and strong SQL, Python and Power BI skills. Reserve 28 active weeks within November 2026–May 2027: 560 hours, comprising 500 planned work hours and 60 contingency hours. This is available project capacity, not a claim about the programme's required credit workload; confirm sufficiency and the submission date with the supervisor.

Work package	Hours	Completion evidence
WP1 · research, scope and contracts	60	Literature matrix; approved RQs; source audit; frozen evaluation protocol.
WP2 · ingestion and reliability	100	Two adapters; revisions/backfills; run ledger; failure and recovery tests.
WP3 · SQL/dbt and data model	70	Documented star model; tested SQL; lineage; measured query optimisation.
WP4 · analysis and BI	90	Five reconciled measures; three report pages; statistical case study; Excel check and decision memo.
WP5 · bounded Fabric demonstration	50	One deployable vertical slice, operational evidence and export; or explicit access limitation.
WP6 · final evaluation	50	E1–E7 results; reviewer feedback or stated fallback; reproducibility check.
WP7 · report, review and portfolio	80	Academic report, revisions, two case studies, demo and handover.
Protected contingency	60	Source issues, integration/review delays and corrections; no default extra features.
Total	560	28 active weeks × 20 hours; writing and research proceed throughout.

Period	Milestone and decision
November 2026	G1: approve topic/RQs, source meaning and access, workload and cloud route. Reconcile repository planning documents with the approved plan.
December 2026	G2: two source extracts, reliable local loading, provenance and replay demonstration.
January 2027	G3: tested SQL/dbt model, CI and first complete source-to-measure slice.
February 2027	G4: three BI pages, all five measures reconciled, draft analytical findings.
March 2027	G5: bounded cloud deployment if available; select final evaluation environment; freeze features and protocol.
April 2027	G6: run final experiments and reviews; finish results/discussion draft.
May 2027	G7: supervisor corrections, reproducibility review, final report and portfolio handover; institutional submission date to be agreed.

Typical week: 8 hours engineering, 6 hours analysis/BI, 4 hours research/writing and 2 hours review/documentation; redistribute by milestone and log actual hours. If delayed, remove stretch work first, then reduce cloud scope. Preserve evaluation and writing; extend the schedule if the required academic workload exceeds capacity.

9. Feasibility, governance and scope control

Risk / trigger	Response and reporting rule
Unverified units or source meaning	Block derived energy/power measures until resolved. Preserve original fields; request clarification through the author if documentation is insufficient. Never relabel MWh as MW.
Insufficient common history	Audit availability before bulk acquisition. Reduce to a justified common period, flag coverage, or agree an alternative source with the supervisor.
Cloud licence/capacity unavailable	Use local core for RQ1/RQ2, retain cloud design and mark practical deployment incomplete. Confirm access and budget before provisioning; export evidence before trial expiry.
Twenty-hour weekly limit exceeded	Use the 60-hour reserve for essential defects and review. Freeze scope at G5. Remove stretch work rather than omit final evaluation.
Reviewer availability	Seek 2–3 reviewers early. If unavailable, use the scripted fallback and limit accessibility claims.
Small data or weak relationship	Treat simple processing and null results as valid. Explain where cloud/Spark add operational capability without claiming a performance necessity.
Security or publication issue	Keep credentials, unapproved data and reviewer details out of Git. Publish synthetic fixtures where raw redistribution is uncertain; document attribution and licence status.

9.1 Academic integrity and information management

Use TAMK's current thesis instructions, reporting guide and required final-report template; this working plan is not a certified institutional template. Confirm the revised topic, programme-specific method and assessment requirements, supervisor, contract/permit process and Wihi milestones [G1–G2]. The original proposal remains available as a separate file.

Prepare a concise data-management plan covering source licences, storage locations, access, backup, retention and disposal. Public energy/weather records and any reviewer contact/feedback data need separate treatment. Agree consent and retention arrangements with the supervisor before recruiting reviewers. Any confidential employer material requires a new scope and permission assessment.

Record AI assistance used in planning, code or text and verify its outputs and citations. The author must understand and defend every analysis and implementation decision. Follow current TAMK disclosure instructions and any restrictions applying to the final abstract or maturity test [G1]. No results, stakeholder participation or supervisor approval may be represented as completed before they occur.

9.2 Explicit scope boundary

Core: batch integration, quality, reproducible SQL modelling, statistical explanation, BI and operational testing. Conditional career objective: a bounded Fabric deployment. Stretch only after G4 and within remaining hours: one forecasting comparison, one extra source or a small additional orchestration example—choose at most one. Fingrid, full infrastructure-as-code, streaming, Kubernetes, multi-cloud deployment, production hydraulic control and generative-AI applications are outside the core.

Track requirements and evidence in GitHub issues and a decision log. Use real peer review when available; otherwise label self-review honestly. Do not present individual project work as experience leading a production engineering team.

10. Deliverables, completion and career use

Output	Definition of done
Academic report	Coherent literature, method, artefact and evaluation; explicit answers to RQ1/RQ2; results, limitations and practical design lessons; supervisor review and TAMK submission requirements addressed.
Engineering release	Versioned source code, contracts, SQL/dbt models, migrations, Docker setup, tests, CI and runbook. Clean-checkout reproduction with a safe fixture; measured replay, correction and recovery.
Analyst release	Power BI file and semantic definitions, five independently reconciled measures, notebook, Excel audit, three findings and a two-page decision memo.
Research evidence pack	Input hashes and licences, environment manifest, experiment scripts/results, query/filter comparisons, failures, timings and anonymised reviewer evidence where available.
Cloud evidence	When implemented: exported pipeline/notebook definitions, Delta schemas, run IDs, access-test evidence and capacity/cost note. Record the access limitation if not implemented.
Recruiter-facing portfolio	One repository with a short quick start, architecture/model diagrams, screenshots and a 5–7 minute demo; two concise case studies emphasising the analyst and engineer contributions respectively.

10.1 Suggested repository organisation

Retain the existing starter and expand deliberately: ingestion/, models/ (dbt SQL and schema documentation), analytics/, powerbi/, fabric/, tests/, experiments/, docs/ and .github/workflows/. Keep raw third-party data and secrets excluded from public version control. Separate generated reports and safe fixtures from private or licence-restricted material.

10.2 Two evidence-based professional narratives

Data analyst narrative: explain the decision question, data limitations, SQL and statistical reasoning, Power BI measures, three supported findings and recommended follow-up. Demonstrate a filter edge case and explain why correlation is insufficient for a causal claim.

Data engineer narrative: explain source contracts, incremental/revision behaviour, modelling decisions, automated tests, failure recovery and deployment trade-offs. Demonstrate a failed run and its recovery, then reproduce a metric from the corrected dataset.

Use measured values only after evaluation. Avoid invented percentage improvements, production-scale claims or implied employment by Helen/FMI. Target roles according to total prior experience and the actual vacancy requirements, including BI developer and analytics engineer roles where the combined evidence is useful.

10.3 Competency sign-off

At G1 record the starting level for Appendix A; at G4 and G7 add an artefact link, test or review result, and one limitation for each competency. Claim independent project competence only when the author can reproduce the work, diagnose a failure and explain the design without following a tutorial. This is an internal portfolio rubric, not a professional certification.

Suggested final report chapters: introduction; literature; research method; requirements and design; implementation; evaluation/results; discussion; conclusions. Keep career-learning evidence in an appendix so that the research argument remains central.

Appendix A1. Analyst and shared competencies

The author already reports strong SQL, Python and Power BI skills. The aim is deeper, demonstrable application. All rows below are core project outcomes; the referenced evidence is planned, not yet completed.

Competency	Practice in this thesis	Evidence of independent completion
Business framing and requirements	Turn the energy-analysis problem into five tasks, metric definitions, personas and prioritised requirements; incorporate a review where available.	Trace each task to a source, model, measure and test; explain one rejected requirement and the scope trade-off.
Advanced analytical SQL	Use CTEs, windows, calendar joins, conditional aggregation, missing-period detection and careful fact joins; inspect an execution plan.	Versioned SQL answers the tasks; edge-case results match fixtures; one optimisation is measured with unchanged results.
Python and reproducible analysis	Reusable analysis functions, explicit types/parsers, environment control and scripted notebook execution.	A clean run regenerates findings and charts; inputs, commit, parameters and dependencies are recorded.
Statistics and critical interpretation	Model historical demand–temperature association, compare calendar adjustment, inspect residuals and assess sensitivity/uncertainty.	Explain seasonality, autocorrelation, confounding and uncertainty; report a weak/negative result honestly; E7 passes.
Power BI, DAX and Power Query	Define relationships, measures, filter context, empty/zero rules and query transformations; parameterise connections and test report interactions.	All five measures pass E5; three report pages answer the tasks; diagnose a deliberately misleading filter/join example.
Excel and independent checking	Create a compact Power Query/PivotTable workbook against an exported Gold sample; document formulas and refresh.	Reconcile one monthly aggregate and one quality count with SQL; make refresh reproducible without manual edits.
Visual communication and decisions	Use appropriate charts, readable units, accessible design, evidence-led narrative and stakeholder feedback.	Two-page memo with three findings and limitations; five-task rubric; concise demo for a nontechnical audience.
Metric governance and domain reasoning	Define units, grain, denominators, lineage, comparison periods and freshness; distinguish measured demand from inferred business effects.	Metric dictionary and provenance trace; defend MW/MWh conversion and DST rules; explain what the data cannot answer.

Learning approach

Use targeted documentation or exercises only when a project task reveals a gap. Keep a short evidence log: skill, baseline, task, result, feedback, remaining weakness. Review it monthly. Count completion through reproducible artefacts and explanation, rather than a list of tools or training hours.

Appendix A2. Engineering and cloud competencies

Competency / priority	Practice in this thesis	Evidence of independent completion
API and batch ingestion · core	Handle CSV/WFS, request boundaries, response validation, rate-aware retries, schema change and acquisition metadata.	Two adapters and repeatable extracts; malformed input, timeout and changed-schema fixtures fail as designed.
Relational design and SQL/dbt · core	Define keys, types, constraints, dimensions, facts, migrations, transformation dependencies and model documentation.	Build and test the model from scratch; explain its grain and revision policy; demonstrate key and relationship checks.
Incremental processing and recovery · core	Implement checkpoints, overlapping backfills, corrected-source handling, atomic publication and a run ledger.	E2/E3 pass; reproduce a correction without duplicate keys and recover without partial analytical output.
Software quality and DataOps · core	Use modular Python, pytest, fixtures, Git branches/PRs, dependency pinning and GitHub Actions checks.	A deliberately broken fixture blocks CI; clean-checkout tests pass; distinguish self-review from external review.
Operations, security and performance · core	Use Docker, shell/CLI operation, stage logs, role separation, backup/restore, EXPLAIN and one measured optimisation.	Runbook, access-denial test, restore into a separate database and performance measurements with environment details.
Cloud orchestration and Delta/PySpark · conditio nal	Implement the bounded Fabric slice, scheduled dependencies, explicit Spark schema, Delta update behaviour and cloud/local result checks.	Actual execution and recovery evidence, exported artefacts and matching sample results. If capacity is unavailable, this outcome remains incomplete.
Cloud governance and cost · conditional	Configure workspace/data access appropriate to the demonstration; inspect actual refresh/capacity behaviour and document cost assumptions.	Show an access test, capacity observation and teardown/export plan. Distinguish manual deployment from automated CD.
Architecture and technical handover · core	Record trade-offs, portability boundaries, lineage and limitations; explain when local SQL is sufficient.	Architecture/model diagrams, at least three substantive decision records and a successful reproduction or documented peer handover.

Assessment levels

Use four levels: 0 = not attempted; 1 = explained or completed with guidance; 2 = independently implemented and reproduced; 3 = tested under failure/edge conditions and critically evaluated. Target level 2 for every core row and level 3 for reliability, metric correctness and one analytical method. Record conditional cloud outcomes separately.

A related analytics engineer application can draw on both appendices, especially SQL/dbt, modelling, metrics, testing and business interpretation. The thesis demonstrates project competence; years of professional responsibility and production scale remain separate claims.

Appendix B. Finnish career relevance and remaining gaps

Employer evidence was checked on 14 October 2026. This is a small purposive sample of four employer-owned pages: two Gofore roles, one Wolt senior role and one Wolt early-career programme. It informs skill selection without estimating market-wide demand or assuming that every listed role is suitable for the author.

Employer evidence	Observed requirement / implication for the plan
Gofore · Data Engineer [J1]	SQL, Python, ETL/ELT, modelling, pipeline ownership and communication; Fabric/Power BI are useful additions. The posting asks for at least five years of experience and Finnish/English. Use it as a skill benchmark, not an entry-level promise.
Gofore · Data & Power BI consultant [J2]	Power BI, DAX, M, SQL, user requirements, modelling and Fabric, alongside communication and Finnish/English. This supports the analyst work and metric governance. It also asks for at least five years of experience.
Wolt · Senior Analytics Engineer [J3]	Advanced SQL, pipelines, data models, business needs and tools including dbt; Helsinki is a listed location. Its five-plus-year expectation makes it a long-term competency reference. The plan adopts transferable skills without copying the entire stack.
Wolt · Data & Analytics internship [J4]	The Helsinki early-career programme describes SQL/Python work for BI developer and analyst interns. It illustrates a different entry route; programme information does not confirm a currently open application or the author's eligibility.

B1. Priorities for applications in Finland

Prepare an English case study and practise explaining the project to technical and business audiences. Assess Finnish proficiency against actual vacancies; a short, reviewed Finnish summary and a domain vocabulary list can support practice but do not establish fluency. Apply at a level consistent with prior experience. Use job descriptions to identify the next gap rather than continually adding technology to the thesis.

The Microsoft PL-300 and DP-700 study guides provide useful self-checklists for analyst and engineering topics [T4–T5]. Certification preparation is optional and outside the 560-hour budget. Neither certification nor this thesis alone guarantees employability.

B2. Skills that need additional experience or separate exercises

Gap	Bounded follow-up after the core thesis
Product experiments and causal inference	A separate synthetic A/B-testing exercise if target roles require it. The heating observations cannot demonstrate a real randomised business experiment.
Other clouds and platforms	Choose one target-role gap later: AWS/GCP, Databricks, Snowflake, Airflow or infrastructure-as-code. Compare concepts before building a second platform.
Production teamwork and operation	Seek internship/work experience, an external code review or an open-source contribution. Solo CI and simulated incidents do not equal production ownership.
Finnish language and interview practice	Continue language practice appropriate to current proficiency and rehearse SQL, modelling, statistics and design explanations using the thesis evidence.

References and source notes

Web sources accessed 14 October 2026. Numbered labels distinguish academic research [A], institutional guidance [G], data publishers [D], technical documentation [T] and employer evidence [J]. Product and recruitment pages may change; recheck before final submission. Convert reference formatting to the programme's required style in the final report.

Academic foundations

[A1] Peffers, K., Tuunanen, T., Rothenberger, M. A. & Chatterjee, S. (2007). A Design Science Research Methodology for Information Systems Research. Journal of Management Information Systems, 24(3), 45–77. https://doi.org/10.2753/MIS0742-1222240302

[A2] Venable, J., Pries-Heje, J. & Baskerville, R. (2016). FEDS: a Framework for Evaluation in Design Science Research. European Journal of Information Systems, 25(1), 77–89. https://doi.org/10.1057/ejis.2014.36

[A3] Wang, R. Y. & Strong, D. M. (1996). Beyond Accuracy: What Data Quality Means to Data Consumers. Journal of Management Information Systems, 12(4), 5–33. Accessible article copy: https://courses.washington.edu/geog482/resource/14_Beyond_Accuracy.pdf

Institutional guidance and datasets

[G1] TAMK. Instructions for thesis. Includes planning, Wihi, integrity and publication guidance; confirm programme-specific requirements. https://opiskelijanopas.tuni.fi/en/tamk/studying-0/thesis/instructions-thesis

[G2] TAMK. TAMK's report guide. Use the current institutional template and reporting instructions for the final thesis. https://opiskelijanopas.tuni.fi/en/tamk/studying-0/tamks-report-guide

[D1] Helen. Open data. The publisher page describes hourly district-heating power; actual file semantics and redistribution terms still require source audit. https://www.helen.fi/en/about-us/helen/open-data

[D2] Finnish Meteorological Institute. FMI WFS services. Source for query structure and service documentation; audit selected station and coverage separately. https://en.ilmatieteenlaitos.fi/open-data-manual-fmi-wfs-services

Local evidence used in this revision

Existing research-plan DOCX and Phase 1 documents; experiments/helen_profile.json; tools/profile_csv.py; compose.yaml; and the preceding local setup checks. The existing audit reports 91,767 rows and the two source column names; it does not confirm their physical meaning, timestamp convention or licence. The author confirmed a balanced career goal, strong SQL/Python/Power BI skills and 20 hours per week.

References continued: technology and career evidence

Technical documentation

[T1] Microsoft Learn. Understand star schema and the importance for Power BI. https://learn.microsoft.com/en-us/power-bi/guidance/star-schema

[T2] dbt Labs. Add data tests to your DAG. Test concepts informing the proposed SQL modelling work. https://docs.getdbt.com/docs/build/data-tests

[T3] Microsoft Learn. Try Microsoft Fabric for free. Trial availability and duration must be rechecked before activation. https://learn.microsoft.com/en-us/fabric/fundamentals/fabric-trial

[T4] Microsoft Learn. Study guide for Exam PL-300: Microsoft Power BI Data Analyst. Optional competency cross-check. https://learn.microsoft.com/en-us/credentials/certifications/resources/study-guides/pl-300

[T5] Microsoft Learn. Study guide for Exam DP-700: Implementing Data Engineering Solutions Using Microsoft Fabric. Optional competency cross-check. https://learn.microsoft.com/en-us/credentials/certifications/resources/study-guides/dp-700

Employer evidence: illustrative sample

[J1] Gofore. Data Engineer. Finland, multiple locations; experienced role. Career page accessible at review time. https://careers.gofore.com/fi/jobs/7543482-data-engineer

[J2] Gofore. Data & Power BI -konsultti. Finland, multiple locations; experienced role. Career page accessible at review time. https://gofore.com/tyopaikat/data-power-bi-konsultti/

[J3] Wolt. Senior Analytics Engineer (Multiple Domains). Helsinki/Berlin/Stockholm. Requirements were available in the employer search-indexed listing; the direct page may render dynamically. https://careers.wolt.com/en/jobs/1/8006892

[J4] Wolt. Early Careers: Data & Analytics internship. Programme description for Helsinki, not confirmation of an open vacancy. https://careers.wolt.com/en/teams/early-careers

Decisions for the next supervisor review

Confirm: revised topic and two RQs; required master's-level contribution and assessment criteria; realistic workload and Wihi dates; source-unit/coverage audit; commissioner and reviewer arrangements; cloud access and the fallback; and the final report/data publication process. This document proposes the work and its evidence; it does not record those approvals as granted.